



DOUGLAS COLLEGE

CSIS 4495 – 002

(Applied Research Project)

Title: EcoTots

(MERN Stack Website for the Exchange of Gently Used kid's Clothing)

Progress Report 2

Submitted To:

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1. Introduction

Since the last progress report, significant advancements have been made in refining the EcoTots platform. The focus has been on improving system security, debugging critical errors, and enhancing platform efficiency. My key contributions during this period include resolving authentication and image upload issues, strengthening security protocols, optimizing performance, and integrating an AI chatbot to assist users.

This report outlines the progress made, challenges encountered, and plans for future improvements.

2. Key Contributions

2.1 Error Handling and Debugging

- Fixed critical Firebase authentication issues preventing user logins.
- Enhanced error handling with detailed messages for failed login attempts and incorrect form submissions.
- Debugged and improved profile update functionality to prevent data loss during edits.
- Resolved Firebase storage permission errors, ensuring seamless image uploads.
- Fixed API request handling errors, improving data flow between the frontend and backend.
- Conducted UI/UX testing to resolve layout inconsistencies across various devices.

2.2 Enhancing Website Security

- Strengthened Firebase authentication with session persistence and token verification.
- Implemented **two-factor authentication (2FA)** for additional login security.
- Improved password encryption and hashing mechanisms to protect user credentials.
- Developed a **role-based access control (RBAC) system** to restrict unauthorized actions.
- Implemented **frontend and backend form validations** to prevent malicious data entries.

- Introduced a user reporting system to flag suspicious activities or fraudulent listings.

2.3 Performance Optimizations

- Optimized **MongoDB database queries** to reduce load times for fetching user listings.
- Implemented **lazy loading** for images to improve page speed and enhance the user experience.
- Reduced redundant API calls, optimizing **server performance and response times**.

2.4 AI Chatbot Integration

- Researched AI chatbot frameworks (Dialogflow, OpenAI API, Rasa) for customer assistance.
- Designed an **initial chatbot flowchart** to assist users with common platform-related inquiries.
- Implemented a **basic chatbot UI** and connected it to a test API for response handling.
- Trained the chatbot on frequently asked user queries, improving its response accuracy.

3. Challenges Faced and Solutions Implemented

Challenge	Solution Implemented
Firebase authentication failures causing login issues	Debugged token handling, improved error messages, and implemented session persistence
Image upload errors due to Firebase storage permissions	Adjusted security rules and improved error handling for better upload reliability
Slow page loading times when retrieving large amounts of data	Optimized MongoDB queries and implemented lazy loading for images
Lack of user trust in security	Introduced two-factor authentication (2FA) and role-based access control (RBAC)

Challenge	Solution Implemented
Users struggling to navigate the platform	Improved UI/UX design, conducted mobile responsiveness testing, and introduced AI chatbot support

4. Repo Check-in Summary

I have committed the following updates to the **GitHub Repository**:

Backend Updates:

- **Controllers:** Updated auth.controller.js, listing.controller.js, user.controller.js to enhance security and error handling.
- **Models:** Refined user.model.js and listing.model.js to improve database efficiency.
- **Routes:** Secured API endpoints in auth.route.js, listing.route.js, user.route.js.
- **Middleware:** Strengthened error handling and authentication mechanisms.

Frontend Updates:

- **Pages:** Improved Home.jsx, Profile.jsx, CreateListing.jsx, SignIn.jsx, and SignUp.jsx.
- **Components:** Added chatbot UI (Chatbot.jsx) and enhanced ImageUploader.jsx.
- **Redux Store:** Updated userSlice.js for better authentication handling.

Miscellaneous Updates:

- **User Research:** Uploaded **local parent surveys** to the **Misc folder**.
- **Security Fixes:** Strengthened Firebase rules and implemented **2FA authentication**.

4. Work Log Summary

Date	Hours	Description of Work Done
Feb 9	1.5	Researched accessibility improvements for better user experience.
Feb 10	1.5	Worked on Firebase storage permissions to ensure better security.
Feb 11	2.0	Fixed profile update functionality, ensuring smooth user edits.

Date	Hours	Description of Work Done
Feb 12	1.5	Implemented frontend validation for secure form submissions.
Feb 13	1.0	Conducted UI/UX testing and resolved layout inconsistencies.
Feb 14	1.5	Researched AI chatbot frameworks for customer assistance.
Feb 15	2.0	Designed an initial chatbot interaction flowchart.
Feb 16	2.5	Explored strategies for integrating the chatbot with the React frontend.
Feb 17	2.5	Implemented a basic chatbot UI and connected it to a test API.
Feb 18	3.0	Debugged and resolved Firebase authentication issues.
Feb 19	1.5	Improved error messages for login failures and incorrect form submissions.
Feb 20	1.0	Performed mobile responsiveness testing and fixed layout issues.
Feb 21	1.5	Optimized database queries to enhance listing load times.
Feb 22	1.5	Improved chatbot response accuracy through additional training.
Feb 23	2.0	Finalized progress report, documenting key challenges and solutions.
Feb 24	2.0	Worked on midterm report and gathered more user feedback.
Feb 25	1.5	Reviewed security vulnerabilities and improved API protection.
Feb 26	2.0	Implemented additional encryption for user password security.
Feb 27	1.5	Integrated Firebase authentication tokens with enhanced security.
Feb 28	2.5	Worked on chatbot NLP training for better automated responses.
Mar 1	2.0	Improved chatbot API connection for real-time interactions.
Mar 2	1.5	Conducted a performance review of Firebase storage optimization.
Mar 3	1.5	Updated Firebase security rules to prevent unauthorized access.
Mar 4	2.0	Enhanced listing filter functionalities for better user search.
Mar 5	1.5	Implemented frontend security measures for form validation.

Date	Hours	Description of Work Done
Mar 6	2.0	Worked on chatbot UI enhancements for better user engagement.
Mar 7	1.5	Debugged mobile layout issues and optimized for various screen sizes.
Mar 8	2.5	Conducted security audits on authentication and payment handling.
Mar 9	1.5	Improved AI chatbot user responses through machine learning updates.
Mar 10	2.0	Researched fraud detection mechanisms for marketplace transactions.
Mar 11	1.5	Implemented reporting system for users to flag suspicious activity.
Mar 12	2.0	Finalized chatbot integration with FAQs and customer support.
Mar 13	1.5	Analyzed error logs and improved exception handling strategies.
Mar 14	2.0	Worked on final security patches and user authentication updates.
Mar 15	1.5	Optimized API calls to reduce server load and improve response time.
Mar 16	2.5	Completed final debugging and testing of chatbot functionality.

5. Future Plans

- Fully integrate the **AI chatbot** into the platform with real-time user assistance.
 - Strengthen **security protocols** with additional fraud detection and monitoring tools.
 - Continue **performance optimizations** to enhance user experience and system reliability.
 - Expand the **user reporting system** to improve platform trust and safety.
 - Conduct further **mobile testing** to ensure responsiveness across all devices.
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6. Conclusion

The work completed in this period has significantly enhanced the EcoTots platform's security, performance, and usability. By resolving major errors, implementing robust authentication mechanisms, and introducing AI-driven customer support, the platform is now more secure and user-friendly. Future efforts will focus on refining the chatbot, strengthening fraud detection, and further optimizing platform efficiency.

This report highlights my contributions and the impact they have had on the development of a secure, scalable, and efficient marketplace for exchanging children's clothing.